



Operating Systems Lab

Course Code: CSE-412

Experiment Title: Introduction to Linux Basic Commands (pwd, ls, clear, history, cat, more, less, nano, vi)

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Experiment Title: Basic Linux Commands and Text Editors.

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Objective of the Experiment

- To understand the purpose and usage of basic Linux commands like pwd, ls, clear, and history.
- To learn how to view file contents using cat, more, and less.
- To differentiate between the nano and vi text editors.

Theory / Background

- Linux is a command-line driven operating system. Common commands allow users to navigate the file system and manage files:
- pwd (Print Working Directory): Shows the user's current location in the directory tree.
- ls (List): Displays the contents of a directory. Options like -l (long format) and -a (all files, including hidden) provide more detail.
- clear: Clears the terminal screen of previous outputs and prompts.
- history: Shows a numbered list of previously executed commands, allowing users to review and reuse them.
- cat, more, less: Used to view file contents. cat displays the entire file at once, while more and less allow for paginated viewing.
- nano vs. vi: nano is a simple, modeless, beginner-friendly editor with on-screen shortcut hints. vi is a powerful, modal editor (Command, Insert, Last-line modes) preferred for its speed and advanced features.

Procedure

- Open a terminal in the Linux environment (e.g., Labex).
- Use the pwd command to print the current working directory.
- Use ls to list files and ls -la for detailed information.
- Use clear to clean the screen, then history to see past commands.
- Create a file named info.txt using touch. Use cat > info.txt to add text, pressing Ctrl+D to save.
- View the file using cat info.txt, more info.txt, and less info.txt. Use nano info.txt to edit and save changes (Ctrl+O, Ctrl+X).

- Practice file operations: create backup.txt, copy content from info.txt using cp, rename a file using mv, and delete a file using rm.

Program Code:

```
pwd
ls -la
clear
history
touch info.txt
cat > info.txt
cat info.txt
more info.txt
less info.txt
nano info.txt
cp info.txt backup.txt
mv backup.txt linux_backup.txt
rm linux_backup.txt
```

Input and Output**Input:**

```
linux is good os
i love linux
we love linux
no one hates linux
we use linux
```

Output:

- `pwd: /home/labex/OS_HW` (example)
- `cat info.txt`: Displays the 5 lines of text.
- `more / less info.txt`: Displays the same text, paginated.
- After `mv backup.txt linux_backup.txt`, `ls` shows `linux_backup.txt`.
- After `rm linux_backup.txt`, the file is deleted.
- `history` lists previously typed commands with numbers.

Analysis / Discussion

The `pwd` command is essential for navigation, preventing users from getting lost in the hierarchical file system. The `ls` command, especially with flags (`-l`, `-a`), provides crucial file metadata. The `clear` command is a simple utility for improving terminal hygiene. The `history` command is a powerful time-saver, enabling users to find and re-run complex commands without retyping them.

Regarding file viewers, `cat` is best for small files, while `more` and `less` are necessary for large files. Between the editors, `nano` is intuitive for quick edits, but `vi` (or `vim`) is significantly more efficient for large-scale coding or system configuration due to its keyboard-driven modal editing, search/replace, and macro capabilities. The practice commands (`cp`, `mv`, `rm`) form the foundation of Linux file management.

Conclusion

This experiment successfully demonstrated the purpose and use of fundamental Linux commands. The key differences between file viewing commands and the `nano` vs. `vi` text editors were understood. The `history` command was identified as a useful tool for command-line productivity and management.